DATA SCIENCE & ANALYTICS · SESSION 09

Clustering with K-Means

Letting the algorithm find the segments — unsupervised learning, start to finish

Asst.Prof.Dr.Chaiwat N. · 2026

2026 · Bachelor Program · 3 hrs · Workshop Format · Hands-on Coding

Session Agenda

01 **Clustering vs Rules**
Why let an algorithm segment

02 **Feature Scaling**
Why you must scale before clustering

03 **K-Means**
The algorithm and how to fit it

04 **Choosing K**
Elbow curve and silhouette score

05 **Profiles & GMM**
Interpreting clusters; soft assignment

06 **Workshop Block**
Cluster the customers and compare to RFM

01

PART 01

Clustering vs Rules

Last session you segmented customers with hand-made rules. Clustering lets the data draw its own boundaries — your first unsupervised machine learning.

Clustering finds groups you didn't define — the data decides the lines

A different kind of learning

Unsupervised	No labels, no right answer to predict. The algorithm finds structure — natural groupings — on its own.
vs rule-based RFM	Rules encode your assumptions; clustering may reveal segments you'd never have drawn by hand.
Same features	We cluster on the same recency, frequency, monetary. Only the method of grouping changes.
Use both	Rules are transparent and explainable; clusters are data-driven. The best segmentation often compares the two.

Source: DS Course, Session 9, 2026

02

PART 02

Feature Scaling

K-Means measures distance, so a variable with a big range dominates. Scaling puts every feature on equal footing — the step beginners skip and regret.

Scale first: K-Means is distance-based, so range matters

Level the playing field

The problem

Monetary runs to thousands; recency to tens. Unscaled, monetary alone decides the clusters — recency is ignored.

StandardScaler

Centers each feature to mean 0 and scales to unit variance, so all three contribute equally to distance.

Always before clustering

Fit the scaler on your features, transform, then cluster the scaled data. Never skip this for K-Means.

PYTHON · SCALING

```
from sklearn.preprocessing import StandardScaler

features = rfm[['recency','frequency','monetary']]

scaler = StandardScaler()
X = scaler.fit_transform(features)

# X now has mean 0, std 1 per column
```

No scaling, no valid K-Means. Every time.

K-Means: assign points to K centers, iterate until stable

The algorithm, in code

The idea

Pick K centers, assign each point to its nearest, move each center to its points' mean, repeat until stable.

Fit & predict

sklearn's fit_predict returns a cluster label per customer in one call — the universal estimator pattern.

Set the seed

random_state fixes the start so your clusters are reproducible. n_init='auto' runs several starts and keeps the best.

PYTHON · K-MEANS

```
from sklearn.cluster import KMeans

km = KMeans(n_clusters=4,
            random_state=42,
            n_init='auto')
rfm['cluster'] = km.fit_predict(X)

rfm['cluster'].value_counts()
```

fit_predict assigns every customer a cluster.

03

PART 03

Choosing K

K-Means needs you to pick the number of clusters. Two diagnostics — the elbow curve and the silhouette score — turn that guess into a defensible choice.

Choosing K: the elbow curve and the silhouette score

How many clusters?

Elbow curve

Plot within-cluster spread (inertia) against K. It always falls; the 'elbow' where it stops dropping fast is a good K.

Silhouette

Scores how well-separated the clusters are, -1 to +1. Higher is better; pick the K with the strongest score.

Judgment too

The math suggests; the business decides. Four clusters you can name beat six the algorithm slightly prefers.

PYTHON · CHOOSING K

```
from sklearn.metrics import silhouette_score

for k in range(2, 8):
    km = KMeans(k, random_state=42,
                n_init='auto').fit(X)
    print(k, km.inertia_.round(),
          silhouette_score(X, km.labels_).round(3))
```

Elbow for spread, silhouette for separation.

04

PART 04

Profiles & GMM

Clusters mean nothing until you describe them — and K-Means' hard edges aren't the only option. GMM assigns soft probabilities instead.

Profile the clusters, then meet GMM's softer alternative

Interpret and extend

Profile

Mean recency, frequency, monetary per cluster names them: 'cluster 0 = recent big spenders = Champions.'

K-Means is hard

Every customer belongs fully to one cluster. But borderline customers don't really — that's the limitation.

GMM is soft

Gaussian Mixture gives each customer a probability of membership in each group — honest about the in-betweeners.

PYTHON · PROFILE & GMM

```
rfm.groupby('cluster')[
    ['recency','frequency','monetary']
].mean().round(1)

from sklearn.mixture import GaussianMixture
gmm = GaussianMixture(4, random_state=42)
rfm['gmm'] = gmm.fit_predict(X)
gmm.predict_proba(X)[:3].round(2)
```

GMM gives membership probabilities, not just a label.

05

PART 05

Rules vs Clusters

Two segmentations of the same customers — compare them, and know when each is the right tool.

Compare rule-based RFM with the K-Means clusters — they should mostly agree

Which segmentation wins?

Cross-tabulate	A crosstab of RFM segment vs cluster shows where they agree and where the algorithm saw something new.
When rules win	When you need transparency and a fixed definition — regulators, stakeholders, repeatable reporting.
When clustering wins	When you want to discover structure you didn't anticipate, or have many features rules can't juggle.
FDA preview	Swap RFM for rolling return × volatility and the same K-Means finds market regimes — bull, bear, sideways.

Source: DS Course, Session 9, 2026

06

Workshop Block

Scale the features, choose K, fit K-Means, profile the clusters, and compare them to last session's RFM segments.

PART 06

By the end of this block, every student has data-driven clusters profiled and compared to RFM

4 Steps · 40 Minutes

Scale	StandardScaler on the RFM features. Confirm mean 0, unit variance before clustering.
Choose K	Run the elbow and silhouette loop. Justify your K with both the diagnostic and a business reason.
Fit & profile	Fit K-Means, then describe each cluster by its mean RFM and give it a name.
Compare	Crosstab clusters against the RFM segments. Where do they agree? What did clustering reveal? Your deliverable.

Source: Workshop, Session 9, 2026

KEY TAKEAWAY

Rules encode what you know. Clustering reveals what you don't.